

H&E-first molecular triage for sarcoma: a reproducible proof of concept on open data

AI in Orthopaedics and MSK 2026 Hackathon, submission document

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Contents

1. Summary	2
2. Results of the baseline notebook	2
3. Payment pathway	4
4. How to run	4
5. Team and contributions	5
Clinical background and implementation strategy	5
1. The clinical problem	5
2. Where the AI sits	5
3. Implementation, step by step	5
4. Integration	6
5. Safety design	6
6. Taiwan	6
7. Diversity and equity	6
Business case: H&E-first molecular triage for sarcoma	6
1. The problem we are solving	6
2. The solution	9
3. Who pays, and why they would	10
4. Payment pathway	12
5. Adoption and roll-out	13
6. Diversity and ethics	13
7. What this submission is and is not	13
8. How much money it saves, and why a government payer pays	13
9. Ethics and regulation, the path we will follow	14
10. Business plan contents at a glance	15
11. Market size	15
12. Competition	16
13. Unit economics	16
14. Team, roles needed and who is in place	16
15. Risks and mitigations	17
16. Milestones and funding	17
17. Development timeline, how fast and what is delivered when	18
18. How much money, line by line	18
19. Where the money comes from, and who is actively giving	20

Business plan in the SBA traditional format	21
1. Executive summary	21
2. Company description	21
3. Market analysis	22
4. Organisation and management	22
5. Service or product line	23
6. Marketing and sales	23
7. Funding request	23
8. Financial projections	24
9. Appendix	24

1. Summary

Problem. Sarcoma subtype decides the operation and the oncology plan, and many subtypes are defined by a gene fusion. The molecular test that names the fusion is slow and centralised: in England urgent solid tumour panels have a 21-day target ([East Genomics](#)), in Taiwan National Health Insurance reimburses next-generation sequencing once per lifetime and only at regional hospitals or above ([Health Promotion Administration](#)).

Solution. A model that reads the routine H&E slide already produced for every biopsy and returns, in minutes, a subtype probability and a recommendation for which genomic test to order first. It does not replace the test. It makes the first order the right one.

Efficiency lever. One H&E slide plus one model call replaces up to three weeks of waiting and one avoidable round of testing per misrouted case.

What is in this submission.

File	Content
technical/01_sarcoma_hne_triage.ipynb	Working baseline on open TCGA-SARC data, 60 patients, out-of-fold AUC 0.82
technical/02_savings_model.ipynb	Transparent savings arithmetic, every input labelled sourced or assumption
medicine/clinical_implementation.md	Clinical background and four-stage implementation strategy
business/business_case.md	Payer, payment pathway, adoption, ethics and regulation, market, competition, unit economics, team, risks, milestones
technical/requirements.txt, technical/fetch_overviews.py	Environment and download helper
group-member-contact/README.md	Team and contributions

2. Results of the baseline notebook

Data. All 600 TCGA-SARC diagnostic slides were listed through the public GDC API. Two subtypes were kept, leiomyosarcoma (LMS) and dedifferentiated liposarcoma (DDLPS), one slide per patient, 30 patients each. Only the lowest pyramid level of each slide was read over HTTP range requests, so the whole dataset is about 60 small images instead of 60 gigabytes ([GDC TCGA-SARC](#)).

Method. Each overview was tiled at 224 px, background removed, tiles passed through a frozen ImageNet ResNet50, features averaged per slide, and a logistic regression trained with 5-fold stratified cross-validation, seed 42, CPU only.

Result. Out-of-fold AUC **0.82**, per-fold 0.61 to 1.00, 42 of 60 patients correct at a 0.5 threshold (Figure 1).

Where it fails. The most confident errors are DDLPS slides dominated by fatty well-differentiated tissue or by large necrotic areas, which a mean over overview tiles cannot resolve. These cases define the next iteration: full-resolution tiles and attention pooling (Figure 2).

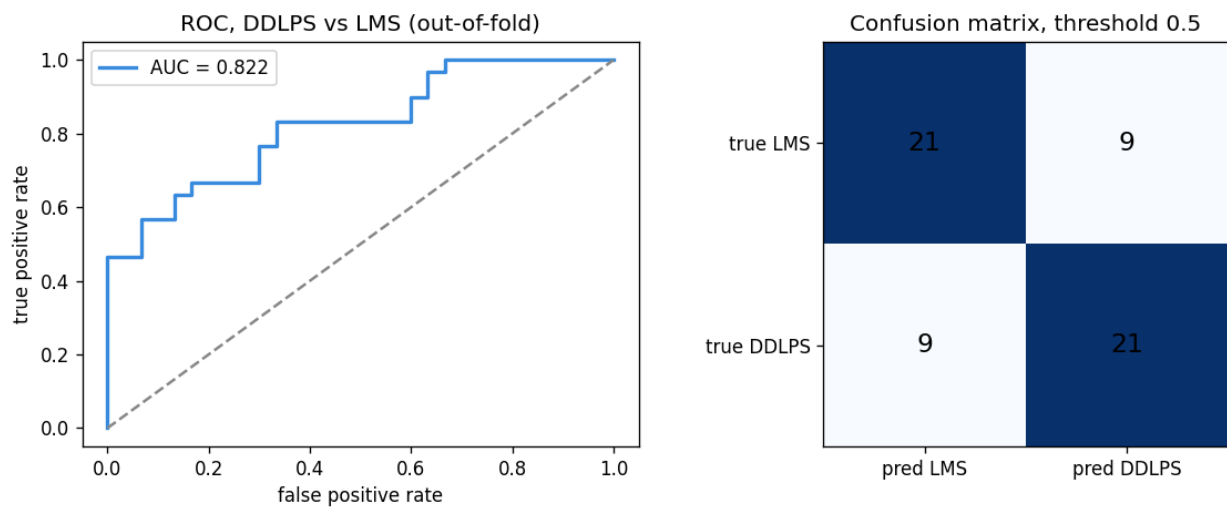


Figure 1: ROC curve and confusion matrix, out-of-fold

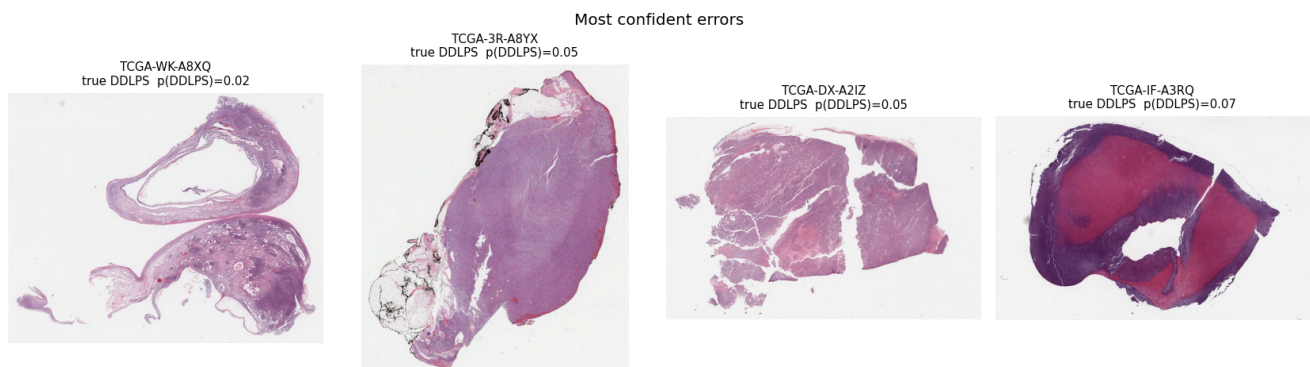


Figure 2: Most confident errors

Limitations, stated plainly. Sixty patients, two subtypes, overview resolution only, a smallest-file selection bias, and a single scanning programme. The fusion-defined subtypes that are the clinical goal have too few open slides today. The production plan and the robustness gate are in the business case.

3. Payment pathway

Figure 3 shows who pays whom along the diagnostic pathway. Grey arrows carry the sample and the report, green arrows carry money.

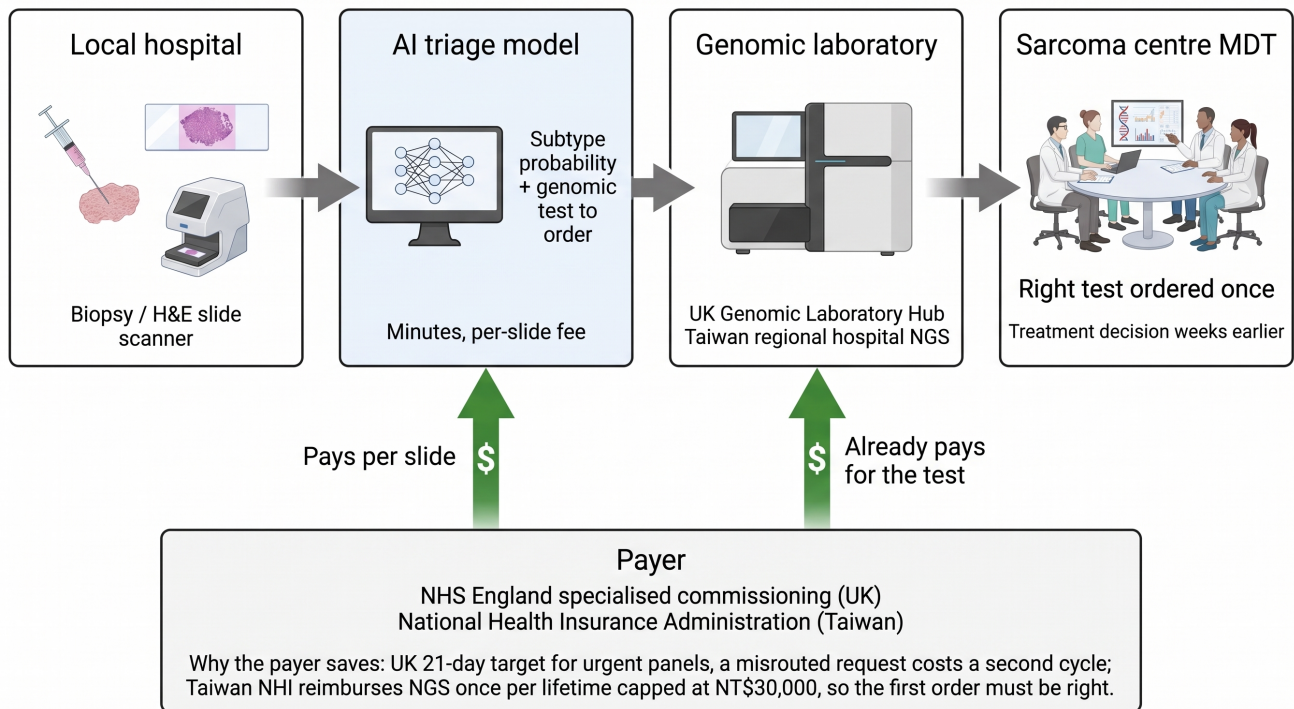


Figure 3: Payment pathway: sample and report in grey, money in green

4. How to run

```
conda create -n sarc python=3.12
pip install -r technical/requirements.txt
cd technical
jupyter lab 01_sarcoma_hne_triage.ipynb # run all, first run downloads ~60 overviews
jupyter lab 02_savings_model.ipynb
```

To see the trained baseline run on a slide, open the browser demo (no GPU, no server beyond a static file server):

```
cd demo
python -m http.server 8765
```

Then open <http://localhost:8765/>, drop an H&E overview image or click one of the six bundled TCGA-SARC slides. The model runs in the browser in a few seconds and shows the subtype probabilities and the panel it would order first. `demo/README.md` explains the files.

Seed 42 is fixed in both notebooks. Features and overviews are cached under `technical/data/` after the first run.

5. Team and contributions

I-Han Cheng, team lead and sole member. Problem definition, data selection, model design, both notebooks, clinical implementation strategy, business case. Partners approached for the production phase, not authors of this submission: UCL sarcoma pathology group.

The clinical implementation strategy and the business case follow on the next pages.

Clinical background and implementation strategy

Prepared 1 October 2026 for the AI in Orthopaedics and MSK 2026 Hackathon. The model is in `technical/01_sarcoma_hne_trial` the payer case in `business/business_case.md`.

1. The clinical problem

Sarcomas are rare cancers of bone and soft tissue, about 5,900 new UK cases a year ([Sarcoma UK](#)). More than 100 subtypes exist and the subtype decides the operation, the margin, and whether chemotherapy is given. Many subtypes are defined by a single genetic event, a gene fusion, rather than by appearance alone ([WHO classification overview](#)).

The diagnostic pathway today:

1. A lump is imaged and biopsied at a local hospital.
2. The H&E slide is read by a general pathologist, who suspects sarcoma.
3. The case is referred to one of the designated sarcoma centres. All suspected bone sarcomas in England must go to such a centre ([RNOH sarcoma unit](#)).
4. The specialist pathologist orders immunohistochemistry and a genomic test through the Genomic Laboratory Hub. The national target for urgent solid tumour panels is 90 per cent reported within 21 days ([East Genomics](#)).
5. The multidisciplinary team meets when the molecular result is back.

Where the time goes: steps 3 and 4. A misrouted or repeated genomic request adds a full cycle.

2. Where the AI sits

Between step 2 and step 3. The H&E slide that already exists is scanned and scored. The output is a probability for each subtype and a recommendation for which genomic test to order first. The pathologist sees the slide before the score. The score never replaces the genomic test and never generates a report on its own.

Efficiency lever: the right genomic test is ordered on day one at the referring hospital, so the sarcoma centre receives a case with the molecular result already in progress.

3. Implementation, step by step

Stage	What happens	Who owns it	Measure of success
Retrospective validation	Model run on archived cases with known fusion status at one sarcoma centre and on the 30-hospital stain-variation dataset	Research team with the centre's pathology department	Per-subtype AUC, per-laboratory AUC, abstention rate
Shadow mode	Model scores live cases, output recorded, not shown to clinicians	One Genomic Laboratory Hub	Misroute rate with and without the model, measured on the same cases

Stage	What happens	Who owns it	Measure of success
Advisory mode	Score shown to the specialist pathologist as a second opinion at the point of test ordering	Sarcoma centre pathology	Days from biopsy to molecular result, repeat-test rate
Referral triage	Score available to the referring general pathologist	Regional pathology network	Proportion of referrals arriving with the correct test already ordered

4. Integration

The model runs inside the laboratory's existing digital pathology system. Slides do not leave the network. Output is a structured field in the laboratory information system alongside the H&E report, with the model version and confidence recorded for audit. Vendors already provide marketplace integration for third-party AI, for example Sectra Amplifier ([Panakeia on Sectra](#)), so no new viewer is needed.

5. Safety design

- The pathologist reads the H&E first. The score appears after the first read, to limit automation bias.
- Low-confidence cases are labelled "no recommendation" and follow the current pathway unchanged.
- Every score is logged with the model version, the scanner and the staining laboratory, so drift at a single site is visible within weeks.
- The model is retrained only under a documented change-control plan, matching the regulator's expectations for adaptive AI ([TFDA predetermined change control guidance](#); the MHRA takes the same approach through the AI Airlock, [GOV.UK](#)).

6. Taiwan

The same tool at a Taiwan medical centre. National Health Insurance reimburses next-generation sequencing once per lifetime, capped at NT\$30,000, and only regional hospitals and above may perform it ([Health Promotion Administration](#)). District hospitals therefore refer without a molecular result. An H&E triage score tells them whether and where to refer, and protects the single reimbursed test from being spent on the wrong panel.

7. Diversity and equity

Sarcoma AI has been trained almost entirely on European and North American slides. Performance is reported separately by laboratory, scanner, sex, age group and ancestry, and the Taiwan cohort exists so that an Asian population is in the validation set from the start, not as an afterthought.

Business case: H&E-first molecular triage for sarcoma

Prepared 1 October 2026 for the AI in Orthopaedics and MSK 2026 Hackathon. The baseline model is in `technical/01_sarcoma_hne_triage.ipynb`, the savings arithmetic in `technical/02_savings_model.ipynb`, the clinical pathway in `medicine/clinical_implementation.md`.

1. The problem we are solving

Why the subtype matters. A sarcoma is not one disease (Figure 4). More than 100 bone and soft tissue subtypes exist, and many are defined by a single pathognomonic gene fusion rather than by what the tumour looks like, for example Ewing sarcoma, synovial sarcoma and myxoid liposarcoma ([Sbaraglia 2021](#)). So the subtype is not a label added at the end. It is the first thing the surgical team has to know, because every later decision depends on it.

What the subtype decides for the surgeon. Three orthopaedic decisions follow directly from the subtype. First, whether to operate now or treat first: Ewing sarcoma and high-grade osteosarcoma receive several cycles of

Sarcoma is not one disease

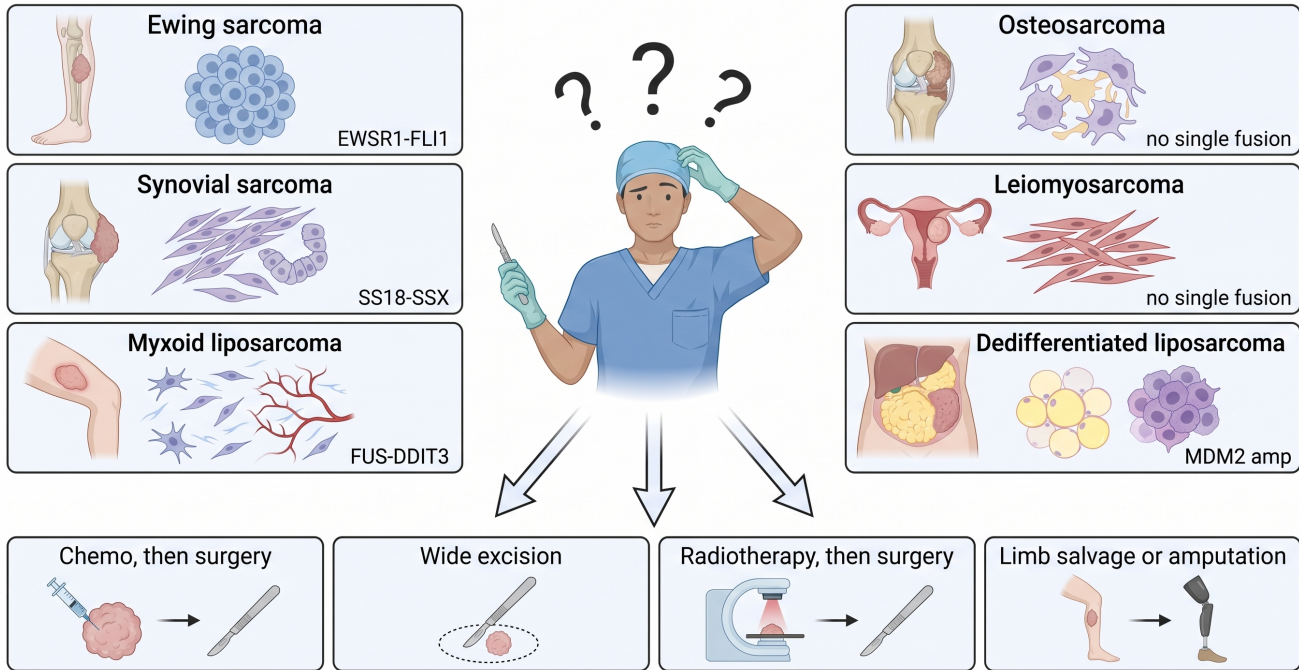


Figure 4: Sarcoma is not one disease: six subtypes with six genetic definitions, one surgeon who has to choose between four different treatment pathways

chemotherapy before any resection, whereas most adult soft tissue sarcomas go to surgery first ([ESMO bone sarcoma guideline](#), [Strauss 2021](#), [UK bone sarcoma guidelines 2024](#)). Second, how wide to cut and whether the limb can be saved: the planned margin, the need for pre-operative radiotherapy, and the choice between limb-salvage reconstruction and amputation are all set by subtype and grade ([UK soft tissue sarcoma guidelines 2024](#)). Third, who should operate and where: in England every suspected bone sarcoma must be referred to a designated centre such as the Royal National Orthopaedic Hospital, which with UCLH sees about 500 new sarcoma patients a year ([RNOH sarcoma unit](#)). Operating on the wrong subtype means the wrong margin, the wrong sequence, or an avoidable amputation.

Why the subtype is an efficiency problem. The subtype is confirmed by a molecular test, and that test is slow and centralised. Figure 6 puts the three points side by side. Here is what it costs when the test is wrong or late, with the numbers.

- **England, the test.** Sarcoma genomic testing runs through seven Genomic Laboratory Hubs, with 54 sarcoma indications in the National Genomic Test Directory ([Birmingham Women's and Children's, genomic testing in sarcomas](#)). The national standard for urgent solid tumour panels is 90 per cent reported within 21 days ([East Genomics, turnaround times](#)). The target is 90 per cent of tests reported within 21 days. In practice about one test in three is late: the only hub that publishes its figure delivers "over two thirds" on time and calls that "slightly above the national average" ([South East Genomics, turnaround times](#)), and NHS England publishes no national figure at all ([Future Health, Keeping pace with cancer, 2024](#)). So a request sent to the wrong panel is not a short delay. It is a second three-week wait at the back of a queue that is already running late.
- **England, the patient.** The delay is not only in the genetic test. The whole cancer pathway in England is already slower than its own target. In March 2026 only 72.8 per cent of patients began their first treatment within 62 days of an urgent referral; the target is 85 per cent ([NHS England Cancer Waiting Times, March 2026](#)). For sarcoma specifically, Sarcoma UK's National Sarcoma Survey found that 30 per cent of patients waited at least six months after first consulting a healthcare professional before an accurate diagnosis, 17 per cent waited more than a year, and 35 per cent saw a healthcare professional more than three times before being referred for tests ([Sarcoma UK, Delays cost lives, 2020](#)). The same report records that delay is associated with higher risk of metastasis and of amputation instead of limb-salvage surgery.

- **England, the surgery.** We know how often a sarcoma is operated on before anyone knows it is a sarcoma. According to PubMed, in 2,201 patients referred to the Royal Orthopaedic Hospital, Birmingham, 402 (18 per cent) had already undergone an unplanned excision elsewhere because the lump was not recognised as a sarcoma; 87 per cent needed a second operation, residual tumour was found in 59 per cent of them, and a positive margin in a high-grade tumour carried a 60 per cent risk of local recurrence even with radiotherapy ([Chandrasekar and colleagues, J Bone Joint Surg Br 2008](#)). So nearly one patient in five has an operation that should not have happened, and more than half of them still have tumour left behind.

The operation that should not happen, and how the model stops it. This is the single most expensive failure in the pathway (Figure 5). A patient has a lump. The surgeon takes it for a benign lipoma and removes it whole. The pathology report then says sarcoma, with tumour at the cut edge. The patient needs a second, wider operation, often radiotherapy, and sometimes an amputation that a planned first operation would have avoided. With the model, the lump is biopsied first, the model reads the H&E and says “sarcoma, refer”, and the first operation is done once, at a specialist centre, with the right margin. One condition must be stated plainly: the model can only help when a biopsy is taken before anything is removed. For a lump that is excised without a biopsy there is no slide to read. The proposal therefore pairs the model with the existing guideline rule, biopsy before excision for any suspicious lump, and measures how often that rule is followed as one of its outcomes.

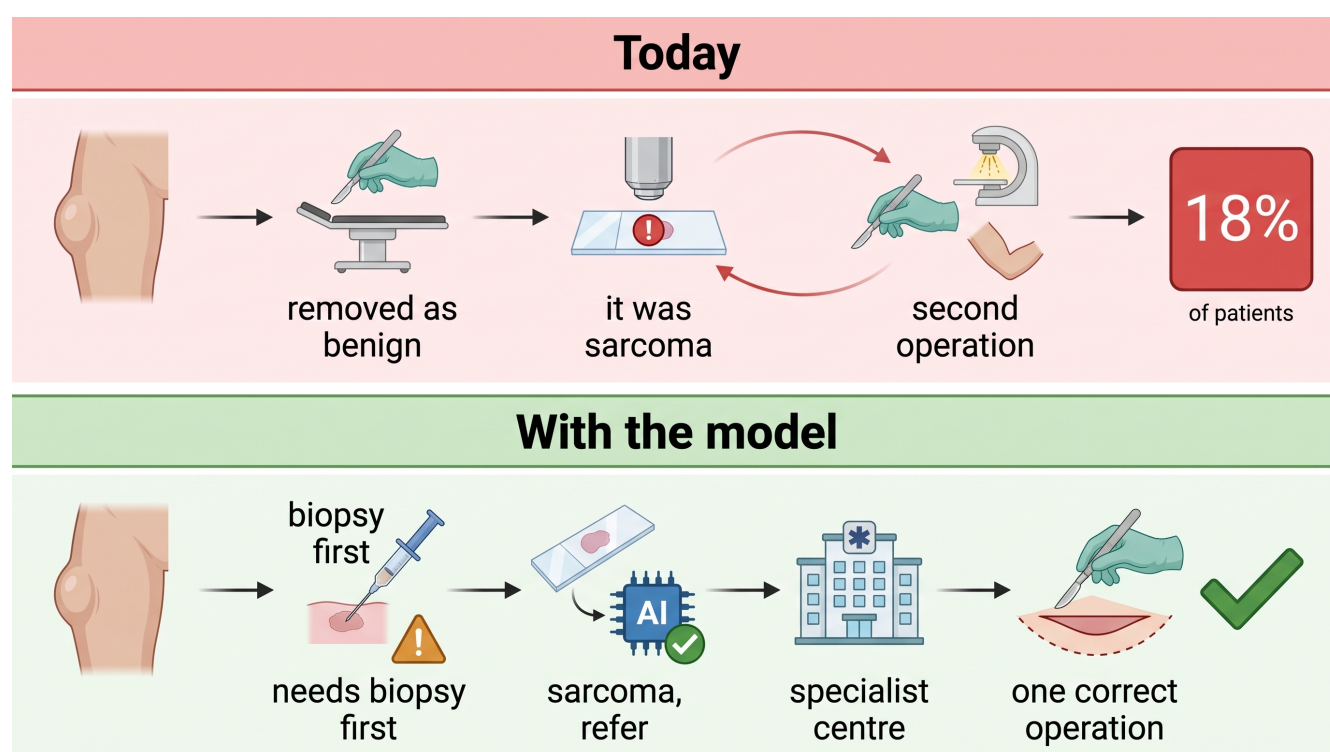


Figure 5: Today: a lump removed as benign turns out to be sarcoma and needs a second operation. With the model: biopsy first, the slide is read, the patient is referred, one correct operation|521x291

- **England, the money.** A 2025 cost study from a large English district general hospital put NHS hospital costs at about £11,200 per patient for an early-stage cancer diagnosis and about £23,800 for a late-stage diagnosis ([BMC Cost Effectiveness and Resource Allocation, 2025](#)). Delayed and incorrect cancer diagnoses also cost NHS Resolution £81.5 million in damages and legal fees between April 2021 and March 2023 ([NHS Resolution review, summarised](#)).
- **Taiwan, the rule.** Reimbursement is set by the National Health Insurance Administration under the National Health Insurance Act's fee schedule. From 1 May 2024 the NHIA pays for next-generation sequencing in 19 cancers at three fixed rates, NT\$10,000 for BRCA, NT\$20,000 for a small panel of up to 100 genes, NT\$30,000 for a large panel, **once per person per cancer type for life**, with an estimated 20,000 patients a year and a budget of about NT\$300 million ([NHIA press release, 2024](#)). Only regional hospitals or above, or hospitals with cancer-care accreditation, listed under the Regulations Governing the Administration of Special Medical Technologies, Examinations, Devices and Equipment as approved laboratory-developed-test sites and running

a molecular tumour board may claim it (NHIA NGS payment Q&A, 2024). District hospitals cannot order it in house.

- **Taiwan, the gap in evidence.** No published figure exists for how many Taiwanese sarcoma patients receive a misrouted first panel, how many referrals are wasted, or how many diagnoses are missed; the scheme is less than two years old and the NHIA has not released utilisation by cancer type. But the rule itself tells us the consequence. If the first panel is the wrong one, the patient's only reimbursed test is gone, and the second one is paid by the patient or not done at all. Producing those numbers is one of the first deliverables of the Taiwan validation arm.

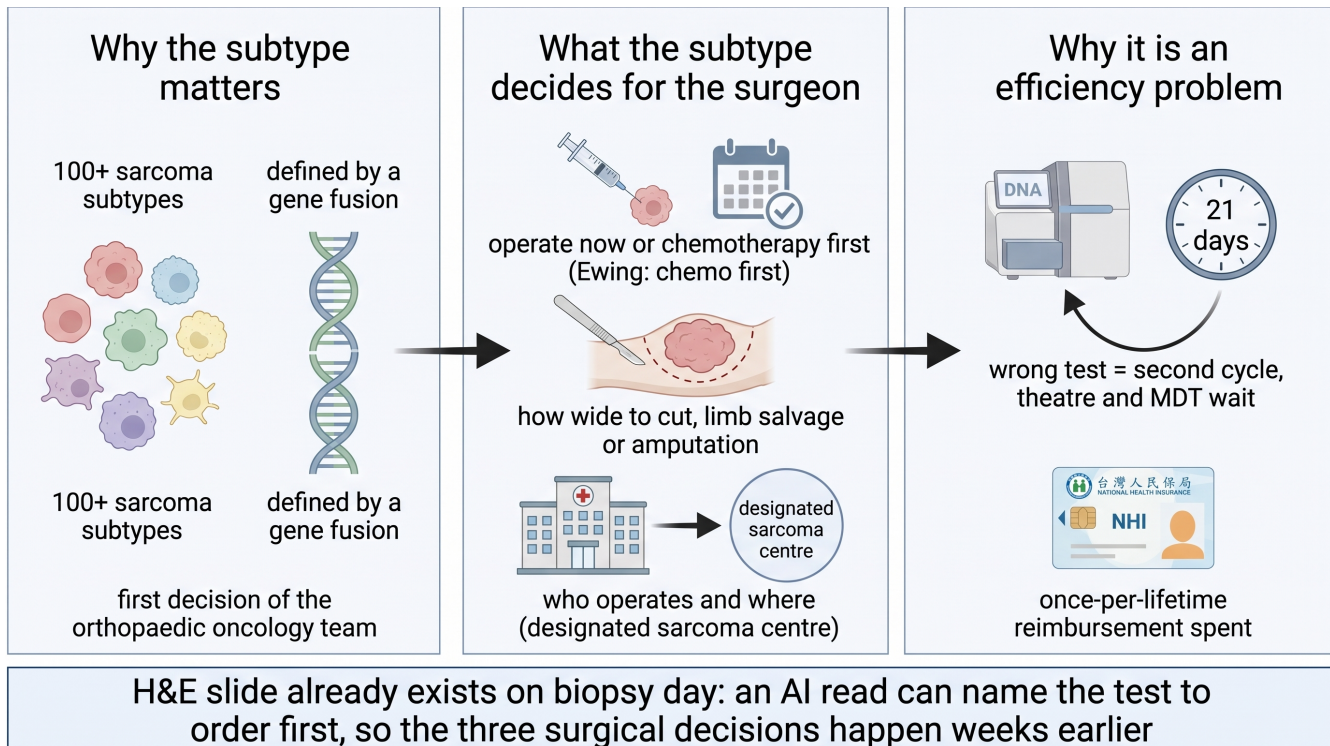


Figure 6: Why the subtype matters, what it decides for the surgeon, and why it is an efficiency problem

The bottleneck is not surgical capacity. It is three things that go wrong before the right operation. A lump is removed before anyone knows it is a sarcoma. A patient is referred to the specialist pathway who did not need it. And the surgical plan waits for weeks between the biopsy and the molecular answer. All three start at the same place, the first H&E slide at the referring hospital. The solution in the next section reads that slide on the day of the biopsy.

2. The solution

The model reads the routine H&E slide, inside the viewer the laboratory already uses, and returns within minutes two probabilities: is this a sarcoma, and which subtype.

the goal of it is to On day one it names the panel to order and how urgently, and flags slides that do not match the requested test.

The orthopaedic oncologist therefore settles at the first meeting, not three weeks later, whether to give chemotherapy first, how wide the margin must be, and whether the limb can be saved.

What is in this submission. Figure 7 shows the two outputs. The notebook here does a first step on the second only, two common subtypes on open TCGA data. The production model adds the first output and the fusion-defined subtypes.

How it is built, and why the data meet the diversity and ethics criteria. Training: open TCGA-SARC slides, then Royal National Orthopaedic Hospital slides under the biobank's existing ethics approval. Robustness: a UK dataset

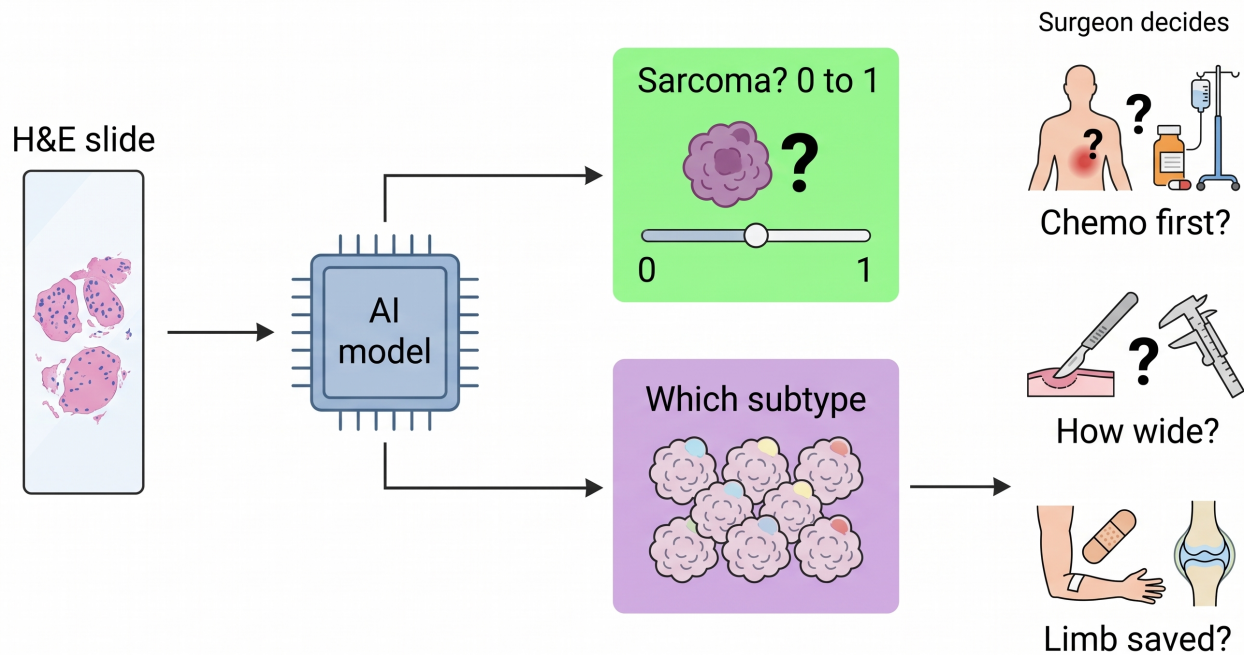


Figure 7: One H&E slide in, two probabilities out, and the three decisions the surgeon can then make

in which one tissue block was stained in 30 hospitals and scanned on 3 scanners. External validation: Taiwanese slides from the National Biobank Consortium, the first Asian cohort in any sarcoma AI. Results are reported by laboratory, scanner, sex, age and population. Patient data never crosses a border; only the model moves.

Where it sits. Figure 8. Today the slide goes to the pathologist, a test is ordered, and a wrong panel repeats the loop; six weeks is common. With the model, one step is added between slide and pathologist. The pathologist reads the slide first, then the score. One correct test is ordered on day one; three weeks in total.

How it plugs in. No new hardware. The model runs as a module in the existing digital pathology viewer, as third-party AI already does through Sectra Amplifier ([example listing](#)). The result is written into the laboratory system next to the H&E report, with the model version for audit. The slide stays inside the hospital network. A working demo of the baseline, upload a slide and read the result, is in `demo/` and takes one command to run (section 4 of the summary).

Figure 9 is a mock-up of that screen. The numbered parts are the only additions to the viewer the pathologist already uses.

Aim. The first genomic request is the right one for every sarcoma patient. Conservatively, about 2,900 patient-days of waiting removed a year in England and about 20 once-only reimbursed tests protected a year in Taiwan (`technical/02_savings_model.ipynb`). Payer and pricing are in sections 3, 8 and 13.

Efficiency lever: **one H&E slide plus one model call replaces up to three weeks of waiting and one avoidable round of testing per misrouted case.**

3. Who pays, and why they would

One payer in each country, two kinds of user, and one channel. Figure 10 shows the money and the use separately.

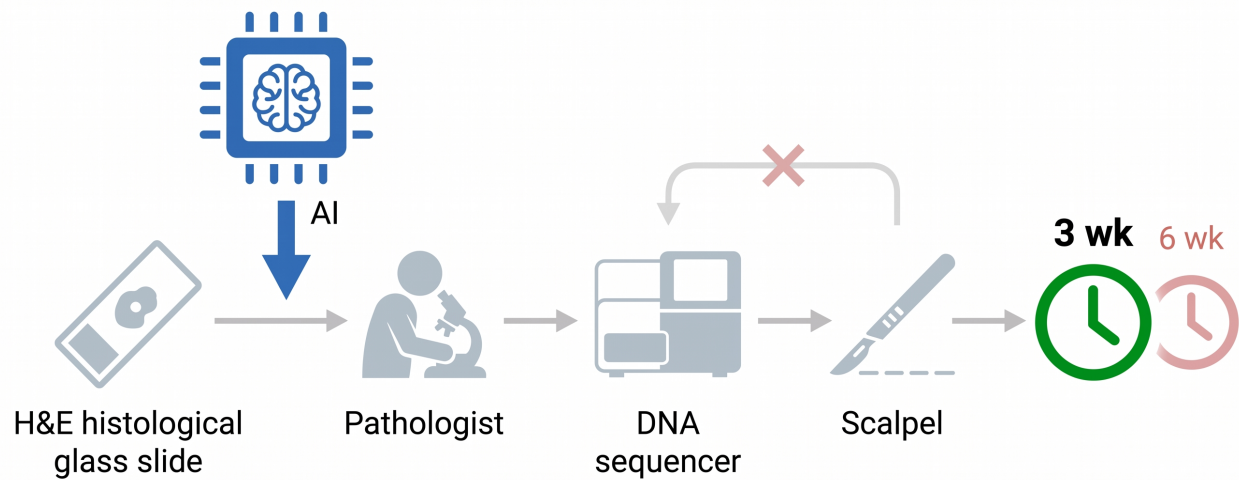


Figure 8: The existing pathway in grey; the AI step inserted in blue; six weeks becomes three

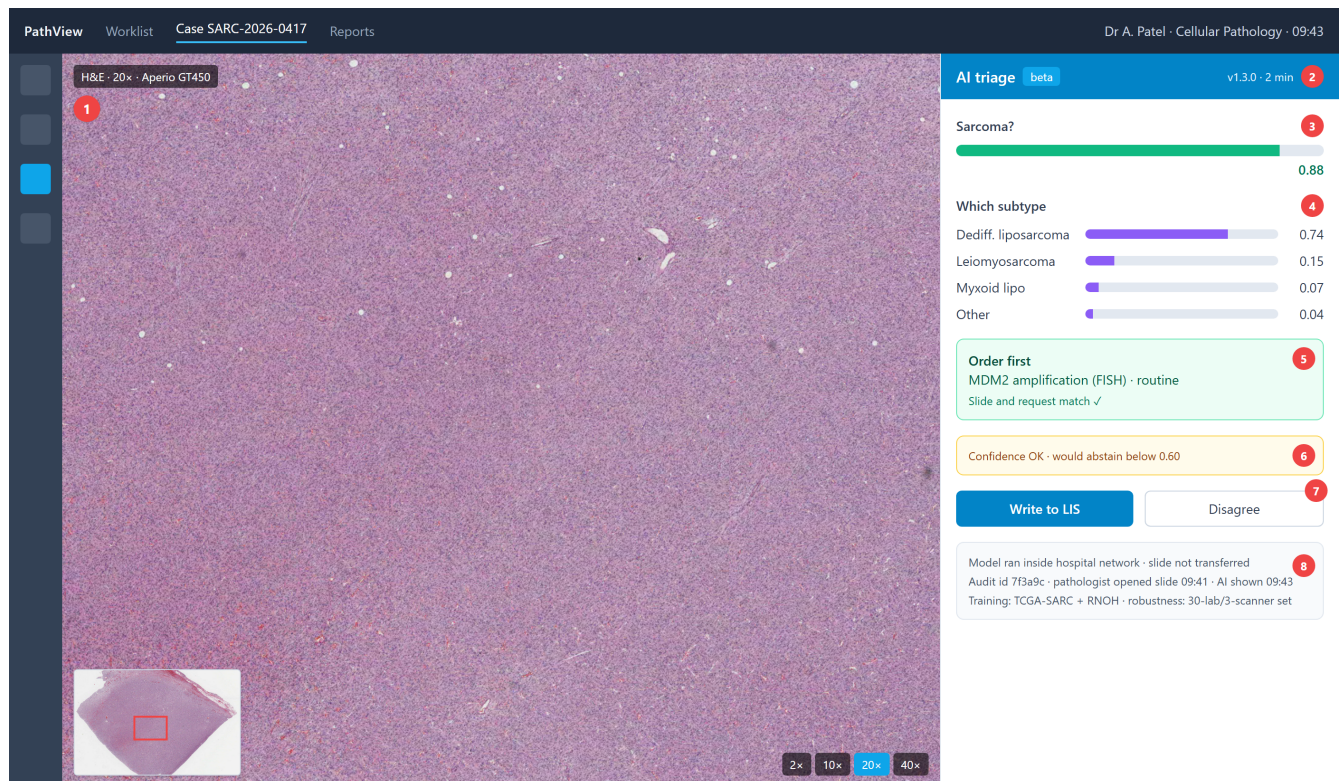


Figure 9: What the pathologist sees: (1) the slide, (2) the AI panel with model version, (3) sarcoma probability, (4) subtype probabilities, (5) the panel to order first and whether slide and request match, (6) the abstain threshold, (7) write to the laboratory system or disagree, (8) audit trail showing the slide never left the network and that the pathologist read first

Who	Role	What they buy	Why it is worth it to them
NHS England, through the Genomic Laboratory Hubs	Primary payer, England	Per-slide licence, or an annual regional subscription, delivered inside the hub's existing digital pathology feed	The hub pays for every genomic test from a central budget. Each misrouted request costs a repeat panel and a second three-week cycle against a 21-day national target. One licence replaces both.
Taiwan National Health Insurance Administration	Primary payer, Taiwan	A reimbursement code for AI-assisted pathology triage	NGS is paid once per patient per lifetime, capped at NT\$30,000. A wrong first panel spends it. Triage protects it.
Specialist sarcoma centres (Royal National Orthopaedic Hospital, Birmingham, Oxford, Newcastle)	User, and the clinicians who ask the hub to buy	Nothing extra; the result arrives with the referral	The orthopaedic team has the subtype before the first multidisciplinary meeting, so the theatre booking, the margin and the limb-salvage decision are made three weeks earlier.
District and regional hospitals (both countries)	User	Nothing extra; the score appears in the viewer they already use	They cannot run genomic tests themselves. The score tells them whether to refer, and stops the lump being removed before anyone knows it is a sarcoma.
Digital pathology marketplaces (Sectra Amplifier, Leica)	Channel	A listed module, paid by revenue share	Scanner vendors do not buy rare-cancer models outright. They list them and take a share, which is how third-party pathology AI already reaches hospitals (Sectra Amplifier listing).

Price against cost. Compute is under £0.10 a slide (section 13). One repeat genomic panel is about £339 ([2017 NHS figure](#)). One unplanned excision followed by re-operation costs the system far more than that; management after an unplanned excision cost 64 per cent more than after a planned one in a Scandinavian series ([EJSO 2020](#)). A per-slide fee in the tens of pounds therefore pays for itself on the first avoided repeat test, and many times over on the first avoided re-operation. The fee used in the financial projections is £40 a slide; it is an assumption until a hub quotes.

4. Payment pathway

The figure (Figure 3) is in the summary section of the submission document and in [business/payment_pathway_biorender.jpg](#) ([editable BioRender version](#)).

Flow: patient biopsy at a local hospital → H&E slide scanned → model call (charged per slide to the laboratory or hub) → triage report → correct genomic test ordered once → result reaches the sarcoma multidisciplinary team.

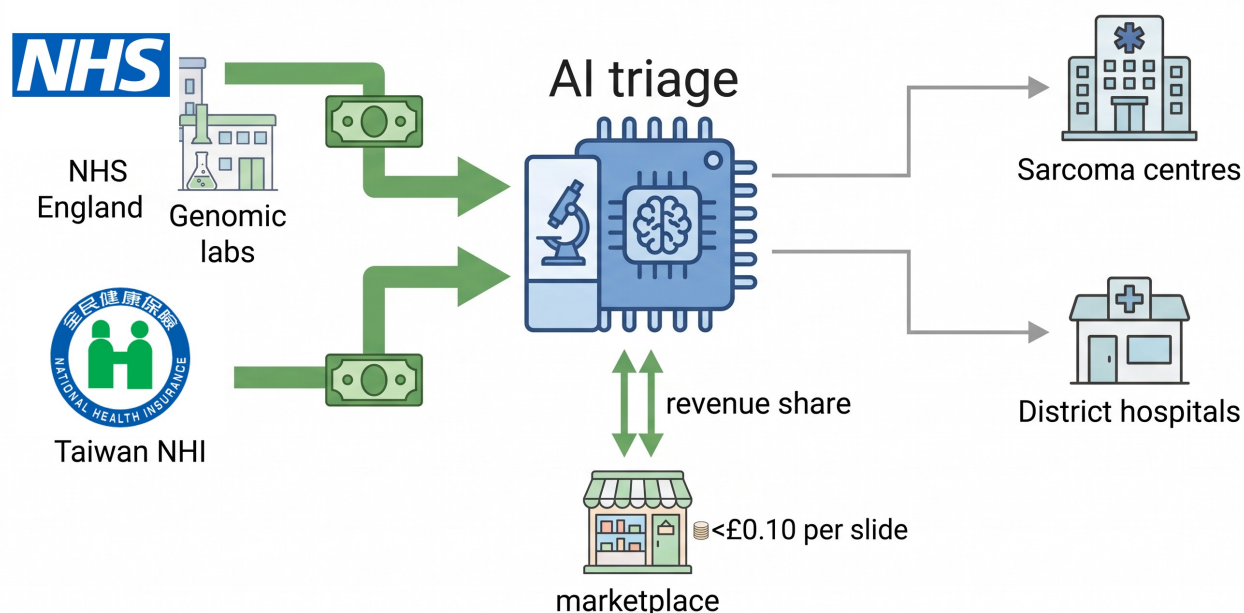


Figure 10: Business model: who pays (green, money in), who uses (grey, result out), and the marketplace channel

5. Adoption and roll-out

1. **Year 1, research use.** Validate on the Royal National Orthopaedic Hospital biobank and on the 30-hospital stain-variation dataset released by the UCL sarcoma group, which shows whether the model survives the change of laboratory and scanner ([Chai, Chen and colleagues, J Pathol Clin Res 2026](#)).
2. **Year 2, shadow deployment** at one Genomic Laboratory Hub, model output recorded but not acted on, measuring how often it would have changed the ordered test.
3. **Year 3, regulated deployment** as a UKCA-marked decision support tool, and a parallel Taiwan pilot at one medical centre with NHI Administration as the reimbursement partner.

6. Diversity and ethics

Performance is reported separately by staining laboratory, scanner and population. The Taiwan arm exists because published sarcoma AI has been trained almost entirely on European and North American slides. Patient-level data are never moved. Only the model moves.

7. What this submission is and is not

The notebook in `technical/` is a proof of concept on open TCGA-SARC data. It distinguishes two common subtypes from slide overviews. It is not the production model. The production model uses a pathology foundation model on full-resolution tiles and is validated on the multi-hospital dataset above.

8. How much money it saves, and why a government payer pays

The arithmetic is in `technical/02_savings_model.ipynb`, with every input labelled SOURCED or ASSUMPTION so a reviewer can change any number and re-run.

Input	Value	Status
New sarcoma cases, UK, per year	about 5,900 (Sarcoma UK)	sourced

Input	Value	Status
Cost of one NHS NGS cancer panel	£339 (2017 figure , historic)	sourced, dated
Urgent solid tumour panel target	90 per cent within 21 days (East Genomics)	sourced
Taiwan NHI NGS reimbursement	once per lifetime, cap NT\$30,000 (HPA summary)	sourced
Share of cases needing a molecular test	0.50	assumption
Share of those tests misrouted or repeated	0.10	assumption
Share of misroutes the triage catches	0.70	assumption
Days saved per caught misroute	14	assumption

With those conservative assumptions the UK direct laboratory saving is about £70,000 a year and about 2,900 patient-days of waiting are removed. The direct saving is small because the test is cheap. The value is in the days, and in Taiwan in protecting a test that can only be reimbursed once.

Why the payer is the government and not the hospital:

1. In England cancer genomic tests are funded centrally by NHS England, so the avoided repeat and the triage licence sit on the same budget ([BWC genomics](#)).
2. The 21-day target is a national performance measure for the Genomic Laboratory Hubs, so a tool that removes second cycles is bought to protect a target.
3. Taiwan NHI pays once per lifetime, so a wrong first order is an unrecoverable public cost.
4. Sarcoma is rare and referral is centralised, so one national licence covers the whole pathway.

The two assumptions a pilot must measure first are the misroute rate and the catch rate. The sensitivity plot in the notebook shows every input moves the result proportionally.

9. Ethics and regulation, the path we will follow

What the tool is in law. Software that triages patient data to influence a diagnostic decision is a medical device in the UK, UKCA-marked under the Medical Devices Regulations 2002, with a UK Approved Body audit for Class IIa and above ([MHRA view](#), [clinician's guide](#)). We plan for Class IIa, decision support that a pathologist confirms, never an autonomous result.

Steps, in order.

1. **Research phase, no device.** Retrospective work on open TCGA data and on biobank material under the biobank's existing ethics (RNOH Biobank, HTA licence 12055, REC 15/YH/0311, [RNOH biobank page](#)). No new patient contact, no new consent needed.
2. **Data protection.** A Data Protection Impact Assessment before any NHS slide leaves the laboratory network. Slides stay in the hub, the model is deployed to the data.
3. **NHS entry criteria.** Digital Technology Assessment Criteria, the NHS baseline covering clinical safety (DCB0129 and DCB0160), data protection, technical assurance, interoperability and accessibility ([NICE and DTAC overview](#)).
4. **Evidence standard.** NICE Evidence Standards Framework for digital health technologies, tier C, which asks for comparative evidence of effectiveness and an economic impact case ([NICE ESF](#)). Notebook 02 is the starting point for the economic case.
5. **Shadow deployment** at one Genomic Laboratory Hub, output recorded, not acted on, with a prospective protocol registered before the first case.
6. **UKCA technical file** and Approved Body audit, then a Taiwan TFDA submission in parallel using the same clinical evidence.

Ethics commitments written into the design.

- Performance reported by staining laboratory, scanner, sex, age group and ancestry, never as one pooled number. The 30-hospital dataset exists to make the laboratory split possible.
- A pathologist sees the H&E first and the model second, so automation bias is limited by workflow, not by a warning label.
- The model can abstain. Low-confidence cases are routed to the full panel, not to a guess.
- No patient-level data crosses a border. The Taiwan arm trains and validates inside Taiwan.
- Patient and public involvement before the shadow deployment, with the sarcoma patient charity as the first contact.

10. Business plan contents at a glance

Section	Question it answers	Where
Problem and customer	Who hurts, how much, today	§1
Solution and product	What it does, what it does not do	§2
Market size	Number of cases, number of laboratories	§11
Revenue model	Per-slide licence to hubs, national licence, OEM to scanner vendors	§3
Competition	Who else does H&E-to-molecular prediction	§12
Unit economics	Cost per inference against price per slide	§13
Go-to-market	Year 1 research, year 2 shadow, year 3 regulated	§5
Regulatory and ethics	Device class, evidence standard, data protection	§9
Team	Clinician, ML lead, pathology partner, regulatory advisor	§14
Risks	Model does not generalise, regulation slips, hubs do not adopt	§15
Milestones and money	What each year costs and who funds it	§16 to §19
Key performance indicators	Misroute rate, days to molecular result, abstention rate, per-site AUC	Listed in notebook 02 §5 and notebook 01 §9

11. Market size

Market	Cases per year	Source
UK, all sarcoma	about 5,900 (700 bone, 5,200 soft tissue)	Sarcoma UK statistics
Taiwan, primary bone cancer	1,238 cases in 2003 to 2010, about 155 a year, age-standardised rate 6.70 per million	Hung and colleagues, Ann Surg Oncol 2014
Taiwan, soft tissue sarcoma	3,843 cases in 2003 to 2011, about 430 a year, age-standardised rate 1.63 per 100,000	Medicine (Baltimore) 2015, Taiwan Cancer Registry

The Taiwan figures are a decade old and are used as a floor. The current annual report of the Taiwan Cancer Registry is the source to update from ([Health Promotion Administration registry reports](#)). Laboratories: seven Genomic Laboratory Hubs in England ([BWC genomics](#)); Taiwan NGS is restricted to regional hospitals and above ([HPA summary](#)).

12. Competition

Nobody markets an H&E-to-fusion-gene product for sarcoma today. The category exists in other cancers, which proves the regulatory route and the buyer, and leaves sarcoma open.

Company	Product	What it predicts from H&E	Regulatory status	Relevance
Owkin	MSIntuit CRC	Microsatellite instability in colorectal cancer, rules out about half of MSS patients at 96 per cent sensitivity	CE-marked (EurekAlert , Nature Communications validation)	Closest business model, a pre-screen that reduces molecular tests
Panakeia	PANProfiler Breast	ER, PR, HER2 status	UKCA and CE (Panakeia)	UK company, same regulatory path we plan
Paige	Prostate Biomarker Suite, Paige Predict	AR amplification, TP53, RB1, PTEN in prostate; 123 biomarkers across 16 tumour types	CE-IVD and UKCA for prostate (Business Wire 2022); Predict launched under Tempus (Stock Titan)	Largest player, no sarcoma model listed
Academic, UCL and RNOH	AI Scope sarcoma diagnosis, UKRI funded	Subtype from pathology plus genomics	Research (UCL news 2023)	Partner, not competitor

Academic literature on sarcoma AI is prognosis-first, not fusion-first, for example survival prediction from H&E with reported AUC 0.97 in one cohort ([PMC review](#)) and margin-aware prognosis in 2025 ([Scientific Reports 2025](#)). A 2025 review frames fusion detection as the next AI target in soft tissue sarcoma genomics ([PubMed 41497157](#)).

13. Unit economics

Per-slide inference cost, computed from public prices. Feature extraction with a pathology foundation model such as UNI takes about 2 to 5 minutes per whole slide on one GPU ([Scientific Reports 2025 retrieval study](#), [ensemble study](#), [PMC](#)). An AWS g5.xlarge with one A10G costs about US\$1.01 an hour on demand ([g5.xlarge pricing](#)).

Item	Value	Basis
GPU time per slide	5 minutes, upper bound	UNI extraction time above
GPU cost per slide	about US\$0.08	$5/60 \times \$1.01$
Storage per slide	1 to 8 GB, about 2 GB compressed	MDPI review of WSI
Storage cost	a 250,000-slide-a-year laboratory spends about US\$90,000 a year	Digital Pathology Association
Volume, UK	about 3,000 slides a year if half of new cases are triaged	\$8

At that volume compute is under US\$300 a year. The cost of the product is regulatory, clinical validation and support, not inference. A per-slide price in the tens of pounds is therefore almost pure margin once the device is certified, and a national licence should be priced against the days saved, not against compute.

14. Team, roles needed and who is in place

Role	Needed for	Status
Clinical lead, orthopaedic and sarcoma Machine learning lead	Problem definition, clinical implementation, MDT access Model, reproducibility, robustness testing	I-Han Cheng, MD, DPhil student NDORMS Oxford, team lead I-Han Cheng for this submission; production model needs a computational pathology partner
Pathology partner	Ground truth, slide access, stain-variation dataset	Planned partner: UCL sarcoma pathology group, contact opened 1 October 2026
Genomics partner	Fusion labels, test-directory knowledge	Planned partner: a Genomic Laboratory Hub
Regulatory and health economics Taiwan clinical partner	UKCA file, NICE ESF economic case Asian validation cohort, NHI pilot	Recruited at grant stage Planned partner: a Taiwan medical centre with NGS accreditation

Contributions for this submission are recorded in `group-member-contact/README.md`.

15. Risks and mitigations

Risk	Likelihood	Mitigation
Model learns stain colour, not biology, and fails at new sites	High, the published sarcoma foundation-model study shows accuracy falling from 0.94 to as low as 0.47 across institutions (Chai and colleagues 2026)	Stain-variation dataset as a gate, stain normalisation, per-site reporting, abstention
Too few open slides for fusion subtypes	Certain today	Start with the two common subtypes, add RNOH and Taiwan biobank material under ethics
Regulatory timeline slips	Medium	Enter as Class IIa decision support, apply to the MHRA AI Airlock phase 3 from April 2026 (GOV.UK AI Airlock)
Hubs do not adopt	Medium	Shadow deployment with a measured misroute rate before any purchase decision
Reimbursement not created in Taiwan	Medium	Pilot under NSTC smart-healthcare funding first, NHI code later

16. Milestones and funding

Year	Milestone	Funding route	Amount and source
1, 2026 to 2027	Production model on RNOH and TCGA slides, robustness on the 30-hospital set, first manuscript	NIHR i4i Product Development Award	No upper limit, typical £0.5 to £1.5 million (NIHR i4i PDA); next round October 2026 (RedKnight note)
1, parallel	Feasibility and business case	SBRI Healthcare phase 1	Up to £100,000 for six months (SBRI Healthcare cancer programme)

Year	Milestone	Funding route	Amount and source
2, 2027 to 2028	Shadow deployment at one hub, DTAC, NICE ESF evidence	Innovate UK Biomedical Catalyst	£25 million pool for industry-led small projects (Innovate UK)
2, parallel	Taiwan validation cohort	NSTC smart-healthcare innovation programme 2024 to 2027 (NSTC)	Submission under the TFDA AI/ML software guidance (TFDA guidance)
3, 2028 to 2029	UKCA technical file, first paid licence	Licence revenue, SBRI phase 2	

17. Development timeline, how fast and what is delivered when

Three years from research prototype to first paid licence. Each quarter has one deliverable that can be checked.

Quarter	Deliverable	Gate to pass
2026 Q4	Hackathon baseline on open data (this submission); first grant application submitted, Cancer Research UK Early Detection and Diagnosis Primer Award	AUC above chance on open data, done (0.82); primer application submitted
2027 Q1	Ethics amendment at the Royal National Orthopaedic Hospital biobank; TCGA full-resolution pipeline with a pathology foundation model	Access letter signed; feature extraction runs on 261 TCGA cases
2027 Q2	First fusion-subtype model (Ewing, synovial, myxoid liposarcoma) on RNOH plus TCGA	Per-subtype AUC reported with confidence intervals
2027 Q3	Robustness gate on the 30-hospital, 3-scanner dataset; stain normalisation on and off	Prediction drift across laboratories below a pre-registered threshold
2027 Q4	Manuscript submitted; Taiwan cohort request through the National Biobank Consortium; shadow-deployment protocol registered	Protocol on a public registry before first case
2028 Q1 to Q2	Shadow deployment at one Genomic Laboratory Hub; DTAC evidence collected	Misroute rate measured with and without the model
2028 Q3 to Q4	Taiwan external validation; NICE Evidence Standards Framework economic case; UKCA technical file drafted	Per-population performance reported
2029 Q1 to Q2	Approved Body audit; MHRA registration; TFDA submission	Certificate issued
2029 Q3 onward	First paid licence to a hub; advisory-mode deployment	Contract signed

18. How much money, line by line

Three-year budget. Sourced lines carry a link; the rest are labelled assumptions and should be replaced by quotes before any grant is submitted.

Line	Year 1	Year 2	Year 3	Basis
Machine learning researcher, 1.0 FTE	£60,000	£62,000	£64,000	Oxford research grade 7, £39,424 to £47,779 salary (Oxford salary scales), plus about 30 per cent on-costs, assumption
Clinical lead time, 0.2 FTE	£15,000	£15,000	£15,000	Assumption, buy-out of clinical sessions
Pathology annotation and slide scanning	£20,000	£10,000	£5,000	Assumption, scanning and pathologist time at RNOH
GPU compute	£2,000	£3,000	£3,000	About US\$0.08 per slide at g5.xlarge rates (AWS g5.xlarge); mostly training, not inference
Taiwan cohort, specimen processing fees	£0	£15,000	£5,000	National Biobank Consortium publishes a fee schedule (NBCT application area); amount is an assumption until quoted
Regulatory consultancy and Approved Body audit	£0	£20,000	£40,000	Third-party UKCA assessment is charged at £200 to £500 an hour and totals from the low thousands upward for a Class IIa software device (UKCA guide); MHRA registration about £300 a year per device category from April 2026 (Casus Consulting); clinical safety officer time is the larger part, assumption
Health economics and NICE ESF evidence	£0	£15,000	£10,000	Assumption, contracted
Patient and public involvement	£3,000	£3,000	£2,000	Assumption, sarcoma charity partnership
Travel, dissemination, open-access fees	£5,000	£7,000	£7,000	Assumption

Line	Year 1	Year 2	Year 3	Basis
Total	£105,000	£150,000	£151,000	About £406,000 over three years

The number to defend is the regulatory line. Everything else is ordinary research cost. If the Approved Body quote comes back higher, the roll-out shifts by a quarter, not the model.

19. Where the money comes from, and who is actively giving

Ranked by fit, with the next deadline where it is published.

Funder and scheme	Amount	Fit	Deadline or status	Source
Sarcoma UK, Open Grant Round 2026	Pilot up to £75,000, Project up to £200,000	High, sarcoma-specific, diagnosis is a named priority	Round closed 10 September 2026, next round 2027	Sarcoma UK open grant round
Sarcoma UK, Improving Sarcoma Diagnosis fund	Per call	Highest, the call names this problem	Rolling, see the Sarcoma UK site	Improving sarcoma diagnosis
Sarcoma UK, Early Career Sarcoma Research Fund	Per call	High, DPhil students eligible	Rolling, see the Sarcoma UK site	Early career fund
Bone Cancer Research Trust, Idea Grant and Ewing sarcoma project call	Project call up to £250,000	High for the Ewing arm, a fusion-defined subtype	2026 calls closed 31 July and 1 June; annual	BCRT for researchers
Cancer Research UK, Early Detection and Diagnosis Primer Award	Up to £100,000, one year	High, pays for exactly the year-1 feasibility work	Rolling rounds	CRUK primer award
NIHR i4i Product Development Award	No upper limit, typically £0.5 to £1.5 million	High for years 2 to 3, covers regulatory and clinical evaluation	Round opens October 2026	NIHR i4i PDA
SBRI Healthcare, NHS Cancer Programme	Phase 1 up to £100,000 for six months	Medium to high, NHS-facing feasibility	Open calls announced on the site	SBRI Healthcare cancer programme
Innovate UK Biomedical Catalyst	£25 million pool, industry-led small projects	Medium, needs an SME or spin-out partner	Annual	Innovate UK
EMERGE RehabTech Pacesetter (seen at ICNR 2026)	Phase 3 awards up to £30,000 for market analysis, grants 1 January to 30 June 2027	Low to medium, rehabilitation technology focus, but the training and mentoring phases are open to UK clinicians and the business-plan coaching is directly useful	Round 1 closed 19 June 2026, Phase 1 workshop 14 July 2026, next round not yet announced; contact EMERGErehabtech@ntu.ac.uk	Medilink Midlands, EMERGE

Funder and scheme	Amount	Fit	Deadline or status	Source
Taiwan NSTC smart-healthcare innovation programme 2024 to 2027	Varies; bilateral AI calls about NT\$5 million a year per project	High for the Taiwan validation arm	Calls listed on the NSTC site	NSTC
MHRA AI Airlock, phase 3	Not a grant, a regulatory sandbox	High for de-risking the device route	From April 2026, three years	GOV.UK AI Airlock

Funding plan in one line. We will start by applying for the Cancer Research UK Early Detection and Diagnosis Primer Award, up to £100,000 for one year, because it is the fastest route to a first grant, it carries the lowest entry barrier for an early-career applicant, and it covers the year-1 budget almost exactly. Sarcoma UK's pilot grant (£75,000) is the parallel application in the next round, and SBRI phase 1 tops up year 1. Years 2 and 3 are covered by one NIHR i4i award. Taiwan is covered by NSTC. The three-year total of about £406,000 fits inside that stack with margin.

Who is actively giving right now. Sarcoma UK and the Bone Cancer Research Trust run annual sarcoma-specific calls and have named diagnosis as a priority. Cancer Research UK's primer award is the fastest route to a first £100,000. The EMERGE Pacesetter seen at ICNR is a business-plan accelerator rather than a research funder, and it is rehabilitation-focused, so it is a place to learn the pitch, not the place to fund the model.

Business plan in the SBA traditional format

Nine sections, following the U.S. Small Business Administration's traditional business plan outline ([SBA, Write your business plan](#)). Each section opens with the SBA's own one-line instruction, then the answer for this project. Numbers are sourced where a link is given; every unsourced number is labelled an assumption. Figures and arithmetic are in `technical/02_savings_model.ipynb` and `business/business_case.md`.

1. Executive summary

SBA: "Briefly tell your reader what your company is and why it will be successful. Include your mission statement, your product or service, and basic information about your company's leadership team, employees, and location."

Mission. Make the first genomic test the right one for every sarcoma patient, by reading the H&E slide that already exists.

Product. Decision-support software that scores a routine H&E whole-slide image in minutes and returns a probability for each fusion-defined sarcoma subtype together with a recommendation for which genomic panel to order first. It is used by the pathologist, never alone.

Why it will succeed. The category already exists and is reimbursed in other cancers: Owkin's MSIntuit for colorectal cancer is CE-marked ([EurekAlert](#)), Panakeia's breast biomarker tool is UKCA-marked ([Panakeia](#)), Paige's prostate suite is CE-IVD and UKCA ([Business Wire](#)). Nobody has built it for sarcoma, where the subtype is molecular by definition and the test is centralised and slow.

Leadership and location. I-Han Cheng, MD, DPhil student at the Nuffield Department of Orthopaedics, Rheumatology and Musculoskeletal Sciences, University of Oxford. One founder at present. Planned partners: the UCL and Royal National Orthopaedic Hospital sarcoma pathology group, one NHS Genomic Laboratory Hub, one Taiwan medical centre. Location: Oxford, with a Taiwan validation arm.

Proof so far. A reproducible baseline on open TCGA-SARC data separates two common subtypes from slide overviews with an out-of-fold AUC of 0.82 on 60 patients (`technical/01_sarcoma_hne_triage.ipynb`).

2. Company description

SBA: "Use your company description to provide detailed information about your company. Go into detail about the problems your business solves."

The problem. Sarcoma has more than 100 subtypes and the subtype decides the operation and the oncology plan. Many subtypes are defined by a single gene fusion ([WHO classification overview](#), [Sbaraglia 2021](#)). The confirming test is slow and centralised. In England the national target for urgent solid tumour panels is 90 per cent within 21 days ([East Genomics](#)). In Taiwan, National Health Insurance reimburses next-generation sequencing once per life-time, capped at NT\$30,000, and only regional hospitals or above may perform it ([Health Promotion Administration](#)). A misrouted or repeated request costs a full cycle in England and an unrecoverable reimbursement in Taiwan.

Who we serve. Genomic laboratories and sarcoma centres that order and run the tests, the referring hospitals that cannot run them, and through them about 5,900 new UK sarcoma patients a year ([Sarcoma UK](#)).

Competitive advantage. Three things nobody else holds together: clinical access to the largest UK sarcoma service through the planned UCL and RNOH partnership, the only public 30-hospital, 3-scanner stain-variation dataset for sarcoma as a robustness gate ([Chai and colleagues 2026](#)), and an Asian validation cohort through Taiwan's National Biobank Consortium, which no published sarcoma AI has.

Legal structure. Not yet incorporated. Research phase runs inside the University of Oxford. The planned route is a university spin-out through Oxford University Innovation at the point of the first NIHR i4i award, with intellectual property assigned under the university's standard terms. This is an assumption to confirm with Oxford University Innovation.

3. Market analysis

SBA: "You'll need a good understanding of your industry outlook and target market. Competitive research will show you what other businesses are doing and what their strengths are."

Industry outlook. Digital pathology is moving from scanning to AI-assisted reporting, with marketplace integration already in clinical use, for example Sectra Amplifier hosting third-party AI ([Panakeia on Sectra](#)). H&E-to-molecular prediction is the fastest-growing product class in that market because it sells a test result without a new test.

Target market, sized.

Segment	Size	Source
UK sarcoma patients a year	about 5,900	Sarcoma UK
UK genomic laboratories for cancer	7 Genomic Laboratory Hubs	BWC genomics
Taiwan sarcoma patients a year	about 585, a decade-old registry floor	Hung 2014 , Medicine 2015
Slides to triage, UK, if half of cases need a molecular test	about 3,000 a year	assumption, see 02_savings_model.ipynb

Competitors. See [business_case.md](#) §12. Owkin, Panakeia and Paige have proven the regulatory route and the buyer in colorectal, breast and prostate cancer. None lists a sarcoma product. Their strength is scale and existing marketplace presence. Our answer is a rare-disease niche they have not entered, with clinical partners they do not have.

4. Organisation and management

SBA: "Tell your reader how your company will be structured and who will run it. Describe the legal structure of your business."

Role	Person or status
Founder, clinical lead, team lead	I-Han Cheng, MD, DPhil student, NDORMS, University of Oxford
Machine learning lead	I-Han Cheng for this submission; a computational pathology post funded at grant stage
Pathology partner	UCL and RNOH sarcoma pathology group, contact opened 1 October 2026

Role	Person or status
Genomics partner	one NHS Genomic Laboratory Hub, to be approached
Regulatory and health economics	contracted at grant stage
Taiwan clinical partner	one medical centre with NGS accreditation, to be approached

Structure. Research project inside the University of Oxford until the first product-development award, then a spin-out limited company with the university and the founder as shareholders. An advisory board of a sarcoma pathologist, a genomic laboratory director and a patient representative is planned before shadow deployment.

5. Service or product line

SBA: "Describe what you sell or what service you offer. Explain how it benefits your customers and what the product lifecycle looks like."

What we sell. A per-slide inference service, deployed inside the laboratory's digital pathology system, returning a structured triage report: subtype probabilities, a recommended first genomic panel, a confidence flag, and an abstain option.

Benefit. The right test ordered once, on day one, at the referring hospital. In England that removes the second 21-day cycle for misrouted cases; in Taiwan it protects the single reimbursed test.

Lifecycle.

1. Research prototype on open data, done (this submission).
2. Production model on RNOH and TCGA slides with a pathology foundation model, 2027.
3. Robustness-gated, stain-normalised version validated on the 30-hospital dataset, 2027.
4. Shadow deployment, then advisory mode, 2028.
5. UKCA-marked Class IIa decision support, 2029, with a documented change-control plan for retraining ([MHRA AI Airlock](#), [TFDA PCCP guidance](#)).

Intellectual property. Model weights, the stain-robust training recipe and the triage reporting format. Foundation models are used under their published licences; this is a dependency to check before commercial use.

6. Marketing and sales

SBA: "Your goal in this section is to describe how you'll attract and retain customers. You'll also describe how a sale will actually happen."

Who buys. The payer is central, not the hospital. In England cancer genomic tests are funded by NHS England, so the budget that pays for the avoided repeat test is the budget that buys the licence ([BWC genomics](#)). In Taiwan the National Health Insurance Administration is the payer.

How a sale happens. Shadow deployment at one Genomic Laboratory Hub produces a measured misroute rate with and without the model. That number, with the economic case in the NICE Evidence Standards Framework format ([NICE ESF](#)), is the sales document. The first contract is a site licence with that hub, the second is a national licence negotiated with NHS England specialised commissioning.

Channels. Direct to hubs and sarcoma centres; marketplace listing on the scanner vendors' platforms (Sectra, Leica, Hamamatsu, 3DHitech) as an OEM module; the sarcoma charity and the British Sarcoma Group meetings for clinical awareness.

Retention. Per-site performance dashboards, drift alerts by laboratory and scanner, and an annual revalidation report delivered as part of the licence.

7. Funding request

SBA: "If you're asking for funding, this is where you'll outline your funding requirements. Your goal is to clearly explain how much funding you'll need."

Amount. About £406,000 over three years, itemised in `business_case.md` §18. Year 1 £105,000, year 2 £150,000, year 3 £151,000.

Type. Grant funding, not equity, for the full three years. The regulatory line (£60,000) is the one to defend; the rest is ordinary research cost.

Sequence. We will start by applying for the Cancer Research UK Early Detection and Diagnosis Primer Award, up to £100,000 for one year ([CRUK](#)), with SBRI Healthcare phase 1 (up to £100,000) in parallel ([SBRI Healthcare](#)). Years 2 and 3 are covered by one NIHR i4i Product Development Award, typically £0.5 to £1.5 million ([NIHR i4i](#)). Taiwan validation is covered by the NSTC smart-healthcare programme ([NSTC](#)). Sarcoma UK and the Bone Cancer Research Trust run annual sarcoma-specific calls that fit years 1 and 2 ([Sarcoma UK](#), [BCRT](#)).

Use of funds. Salaries and clinical time 63 per cent, regulatory and health economics 21 per cent, data and compute 9 per cent, dissemination and patient involvement 7 per cent.

8. Financial projections

SBA: "Supplement your funding request with financial projections. Your goal is to convince the reader that your business is stable and will be a success."

All revenue figures below are assumptions, because no price has been agreed with any buyer. They are shown so the break-even logic can be checked and replaced.

Item	Value	Status
Price per slide to a hub	£40	assumption
Variable cost per slide	about £0.10 (GPU US\$0.08 at g5.xlarge rates, AWS)	sourced
UK slides triaged a year at full adoption	about 3,000	assumption, half of new cases
UK revenue a year at full adoption	about £120,000	derived
National licence alternative	priced against patient-days saved, about 2,900 a year in the conservative model	derived, <code>02_savings_model.ipynb</code>
Taiwan revenue a year, 300 slides at NT\$1,500	about NT\$450,000, about £11,000	assumption

Break-even. With fixed costs of about £150,000 a year in years 2 and 3 and a contribution of about £40 per slide, break-even on per-slide pricing needs about 3,750 slides a year, more than the whole UK sarcoma volume. The business therefore does not break even on UK per-slide fees alone. It breaks even on one of three routes: a national licence priced on days saved rather than slides, an OEM module sold through scanner vendors across several countries, or extension to the next fusion-defined tumour types (paediatric and gastrointestinal stromal tumours) that reuse the same pipeline. The three-year plan is grant-funded precisely so that the first licence is signed with evidence in hand, not with a price guess.

Cash-flow shape. Grants in, no revenue, years 1 to 2. First licence revenue year 3 Q3. The projection assumes the i4i award is secured; if it is not, year 2 compresses to the shadow deployment alone and the regulatory file slips a year.

9. Appendix

SBA: an optional section for supporting documents.

- `technical/01_sarcoma_hne_triage.ipynb`, baseline model, reproducible.
- `technical/02_savings_model.ipynb`, savings arithmetic and sensitivity.
- `medicine/clinical_implementation.md`, clinical pathway and four-stage implementation.
- `business/business_case.md`, §1 to §19, the full narrative with sources.
- `business/payment_pathway_biorender.jpg`, payment pathway figure.
- `group-member-contact/README.md`, team and contributions.